Aplicações do Amplificador Operacional

Eletrónica

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DEIS – ISEC – PT

23/12/2022

Na realimentação negativa

$$G_{MA} = 10^5$$

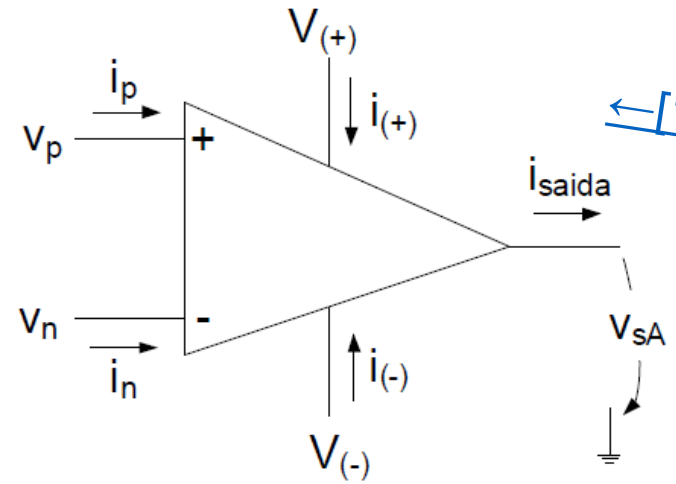
condição de curto-circuito virtual

$$(v_p - v_n) \cong 0 \quad \Rightarrow \quad v_p \cong v_n$$

$$R_{entrada} = 2\,\text{M}\Omega = +2 \times 10^6\ \Omega$$

$$i_p \cong i_n \cong 0$$

Figura 5.17. Função transferência do *amp-op*



←[1]→

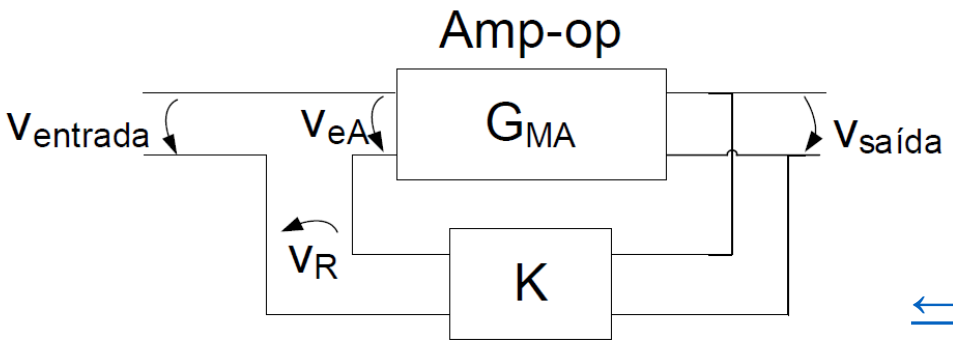
condições na análise de circuitos

$v_p \cong v_n$.

$i_p \cong i_n \cong 0$.

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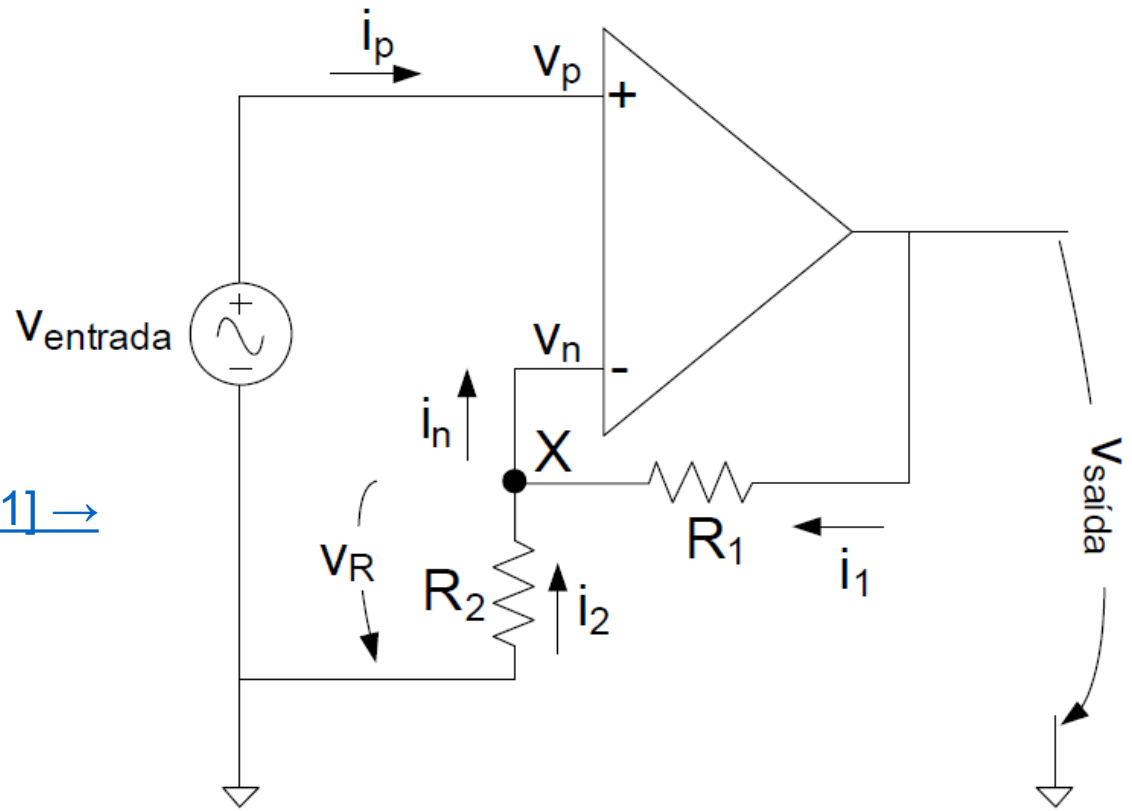
Figura 6.1. Realimentação negativa



$$G_{MF} = \frac{v_{saída}}{v_{entrada}} = \frac{G_{MA}}{K \times G_{MA} + 1}$$

←[1] →

Figura 6.4. Esquema elétrico de um circuito amplificador não inversor



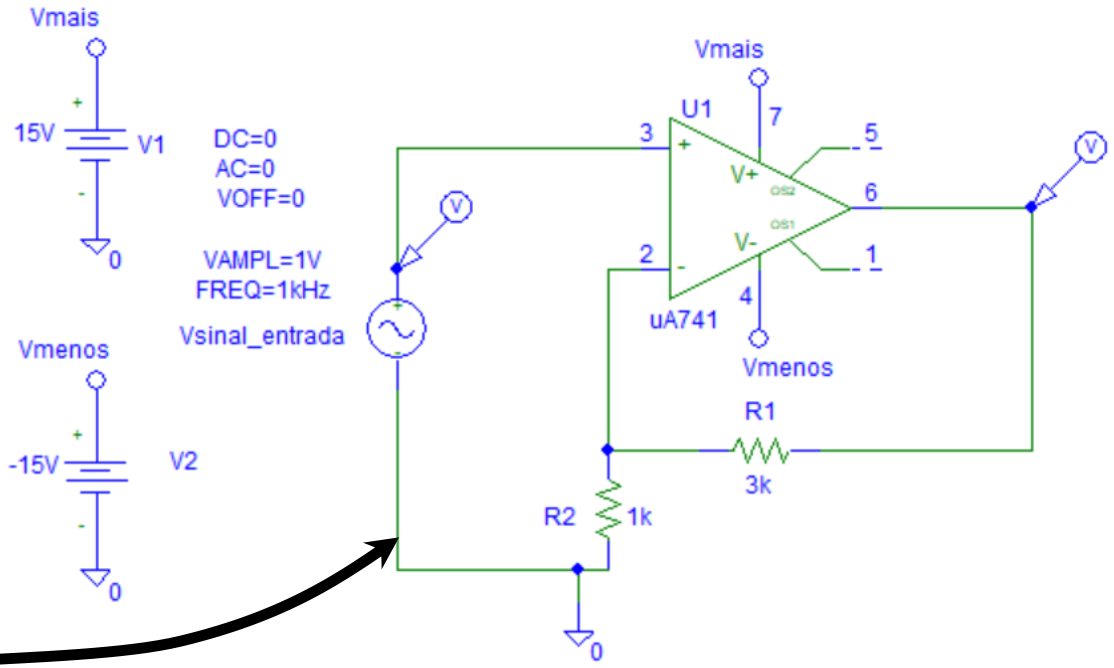
[1] →

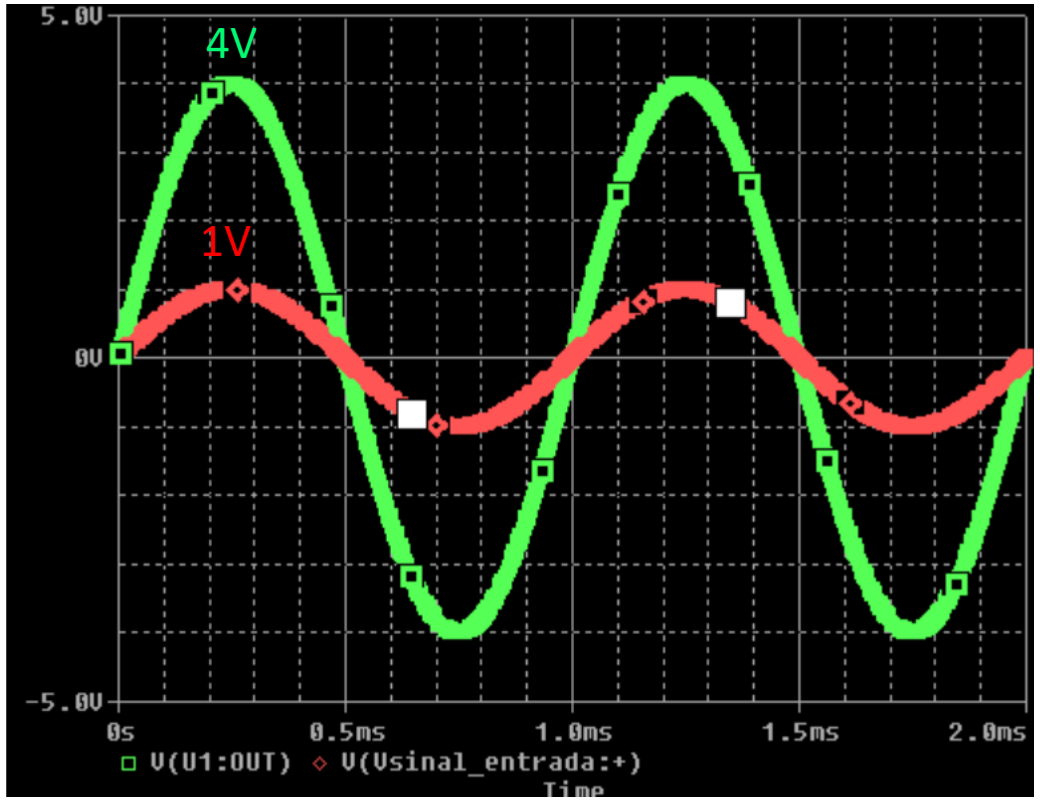
$$G_{MF} = \frac{v_{saída}}{v_{entrada}} = \left(\frac{R_1}{R_2} + 1 \right) \qquad (6.9)$$

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Amplificador não inversor

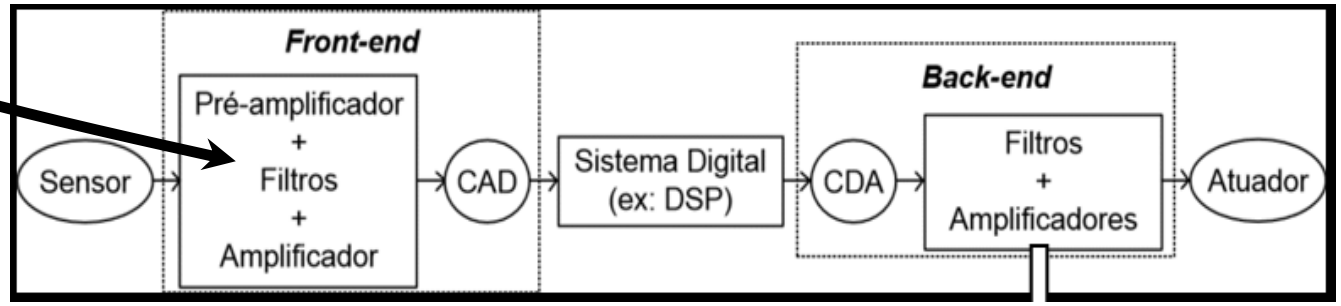
[1] Acácio Amaral, "Eletrónica Aplicada", Edições Sílabo, 2021.





$$Sinal_{saida} = Sinal_{entrada} \times \left(\frac{3000}{1000} + 1\right) = Sinal_{entrada} \times 4$$

[1] →

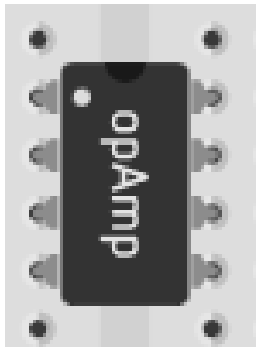


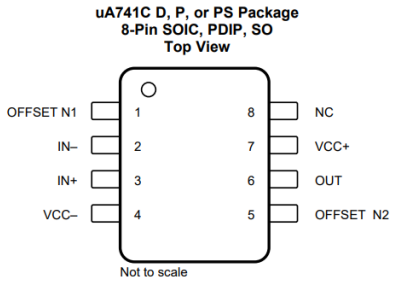
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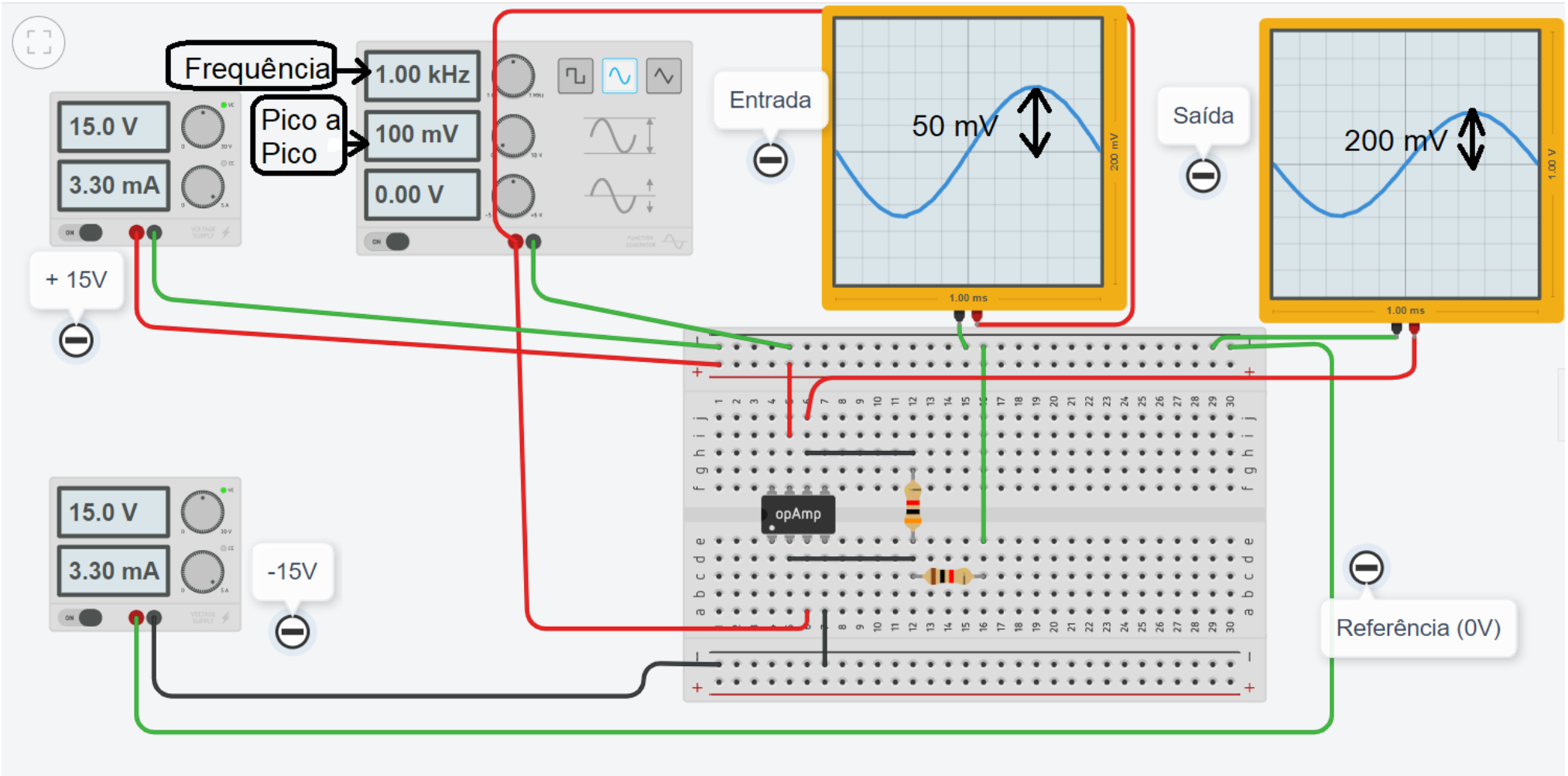
Simulação tinkercad:

Amplificador não inversor

https://www.tinkercad.com/things/6Rn3FpMbclW?sharecode=efrllODqsp-5LElgC21VRrkeECSC9zTWQDBQ11GVPgA





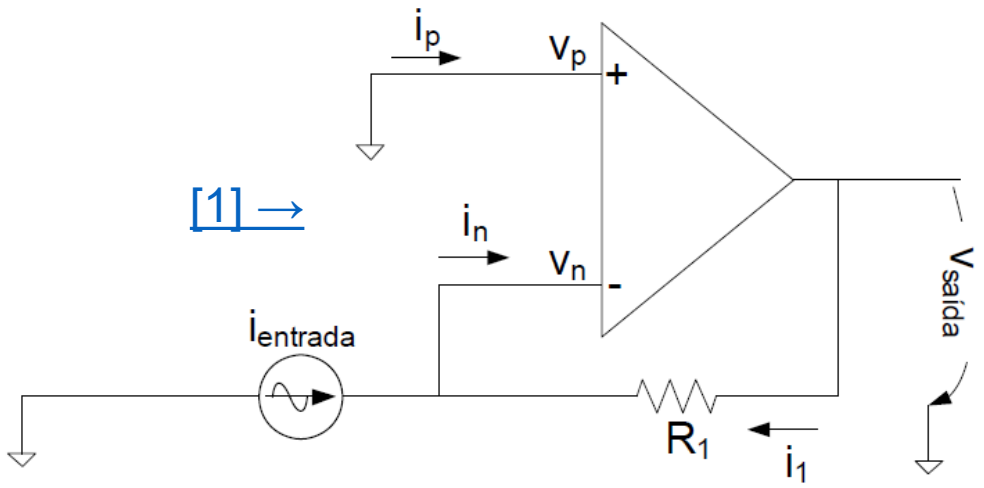


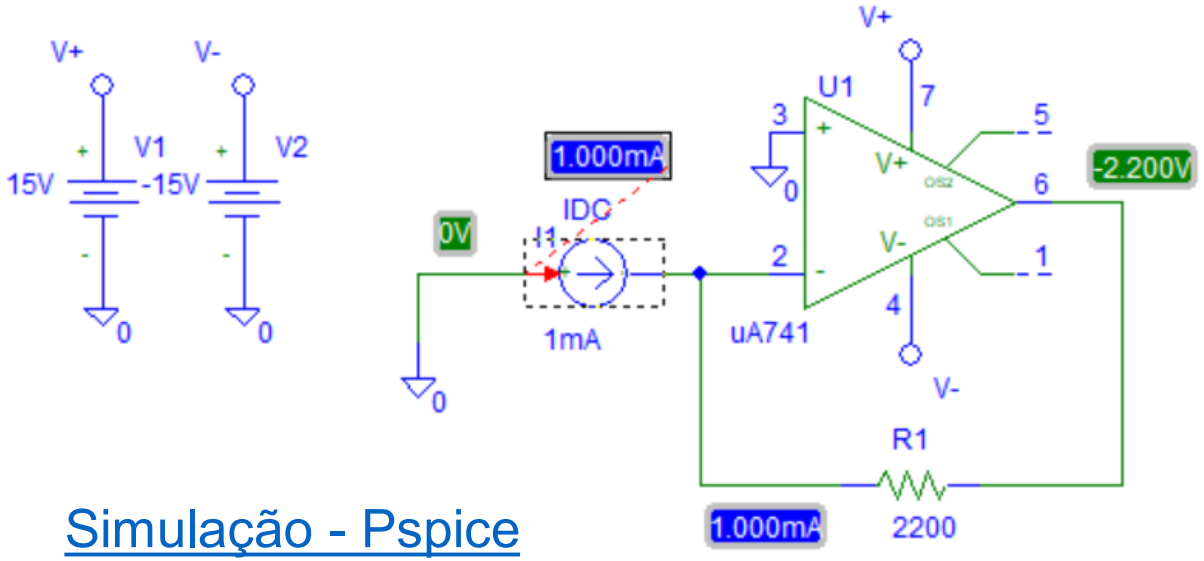
$$Sinal_{saida} = Sinal_{entrada} \times \left(\frac{3000}{1000} + 1\right) = Sinal_{entrada} \times 4$$

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[1] Acácio Amaral, “Eletrónica Aplicada”, Edições Sílabo, 2021.

Figura 6.8. Esquema elétrico de um conversor corrente-tensão





Simulação - Pspice

$$V_{saida} = i_{\ entrada} \times (-R_1)$$

Aplicações



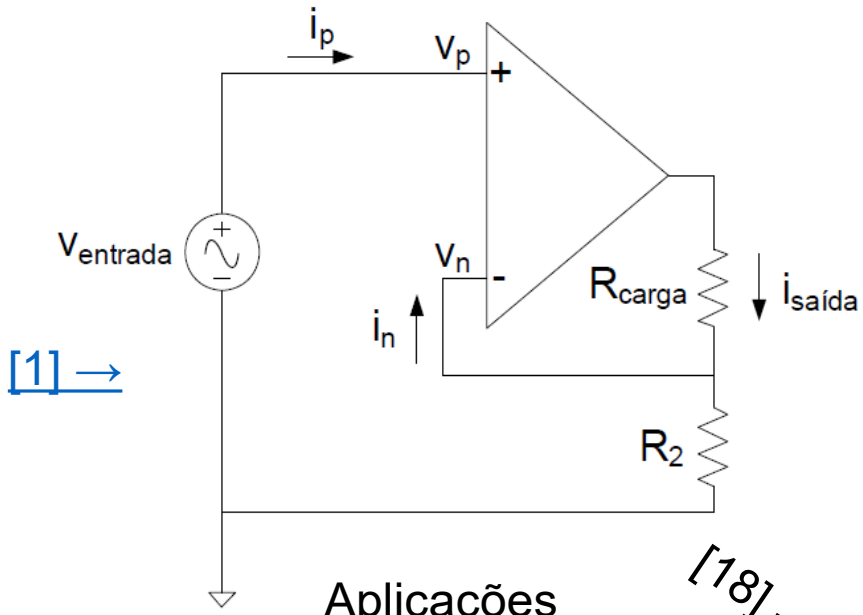






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Figura 6.13. Esquema elétrico de um conversor tensão-corrente



[1] →

$$i_{saida} = \frac{v_{entrada}}{R_2}$$

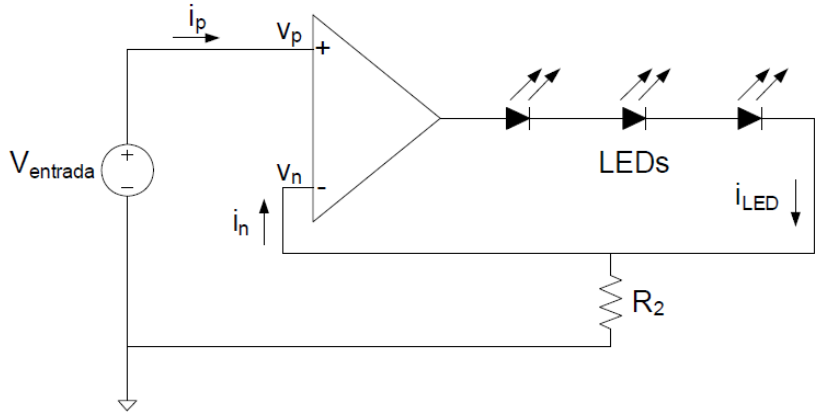
Aplicações [18] →







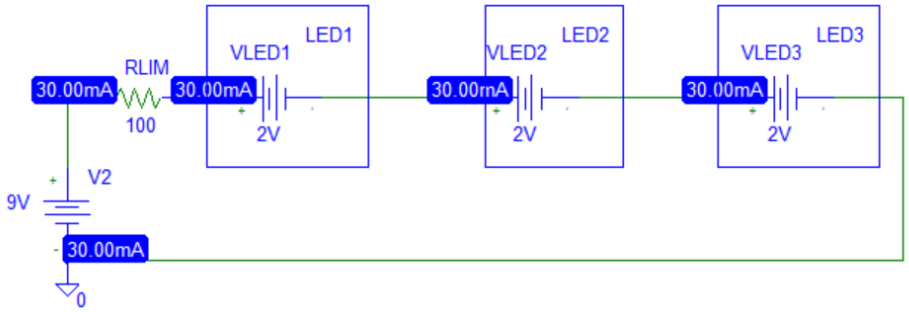
Figura 6.15. Circuito conversor tensão-corrente – utilizado num circuito de iluminação com LED

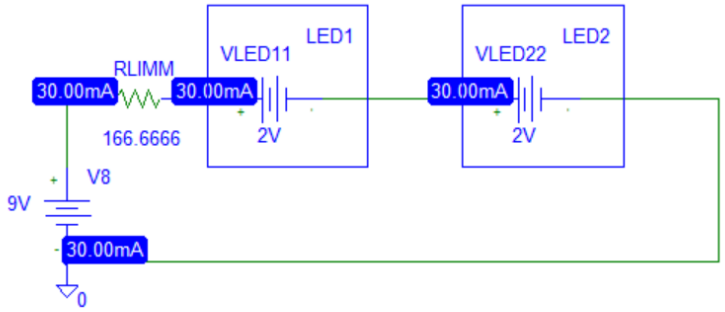


←[1]



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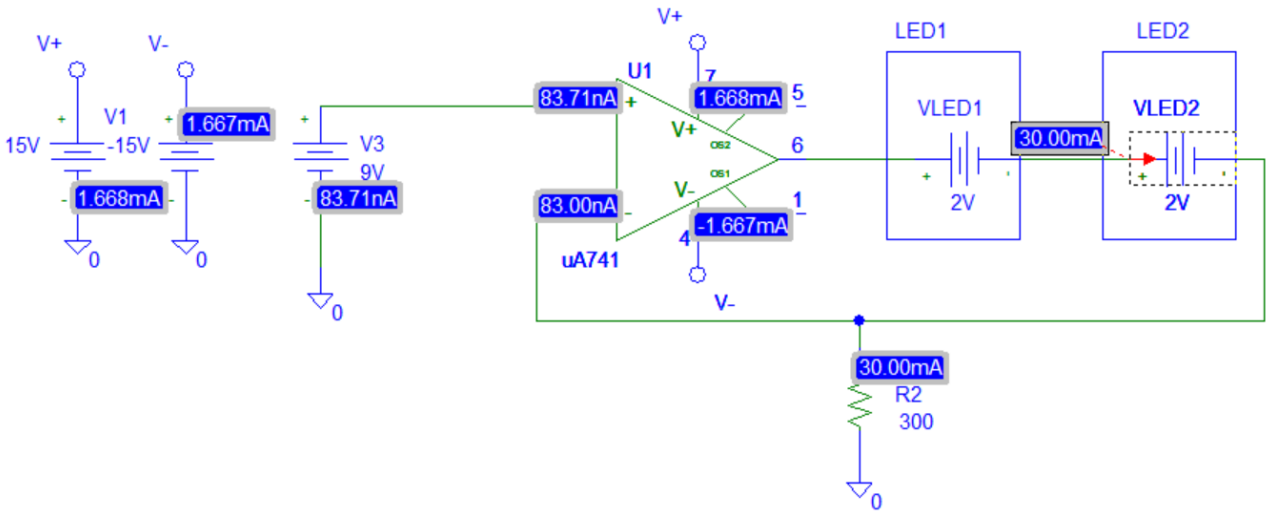
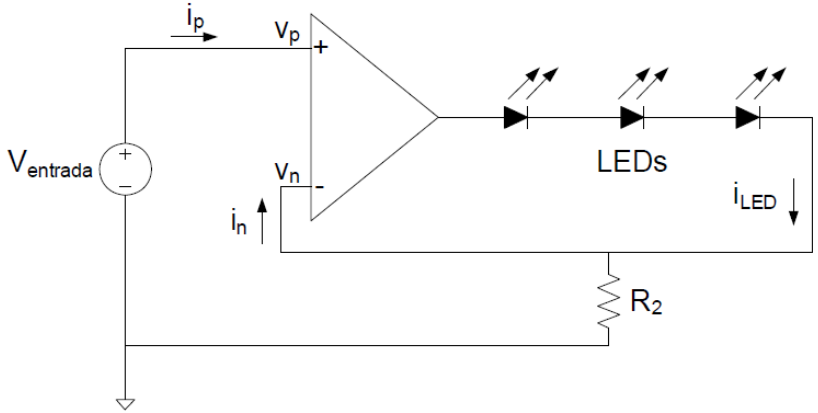
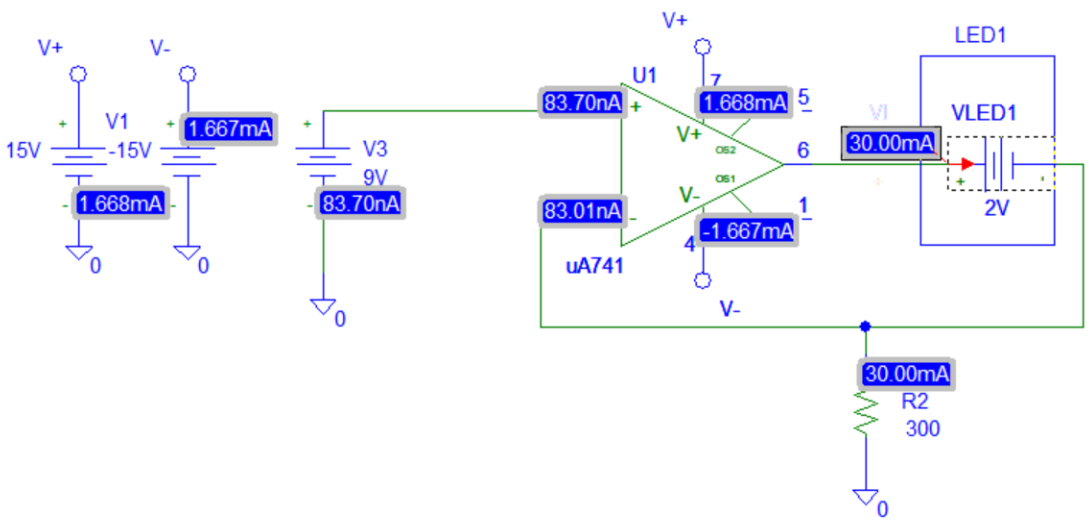


Figura 6.15. Circuito conversor tensão-corrente – utilizado num circuito de iluminação com LED

[1] →



$$G_{MFTC} = \frac{i_{saída}}{v_{entrada}} = (R_2)^{-1} \qquad (6.24)$$



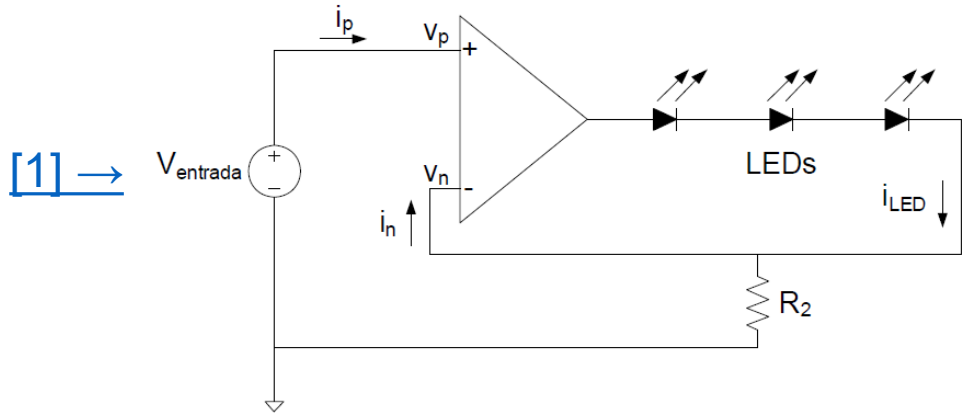
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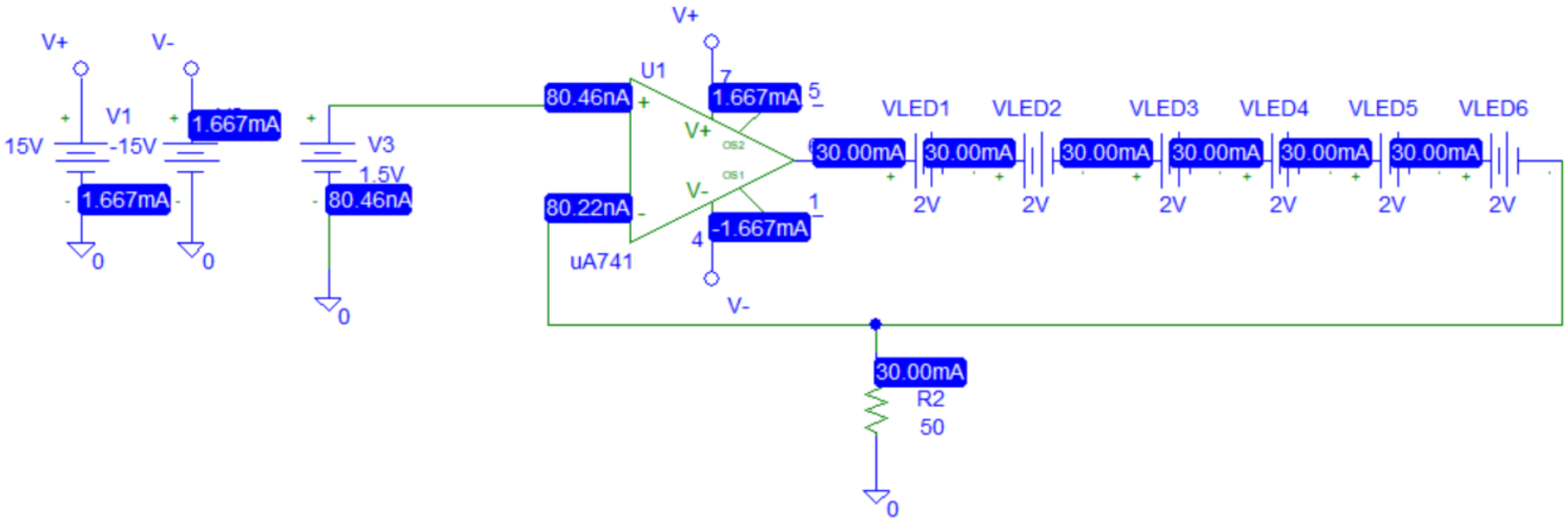
LEDs não acendem se ligados diretamente a uma pilha de 1.5 Volts

(tensão de arranque = 2 Volts)

$$G_{MFTC} = \frac{i_{saída}}{v_{entrada}} = (R_2)^{-1} \quad (6.24)$$

Figura 6.15. Circuito conversor tensão-corrente – utilizado num circuito de iluminação com LED

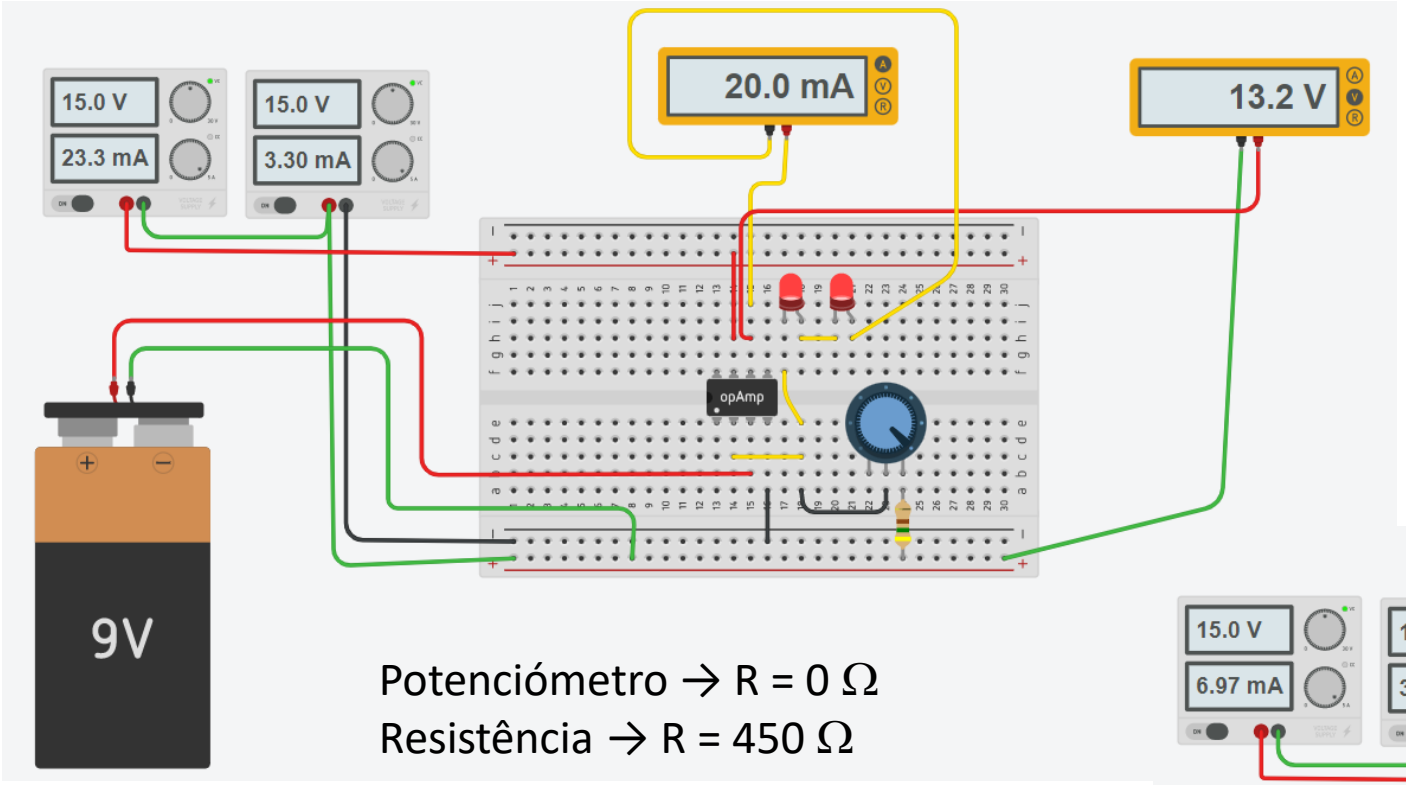




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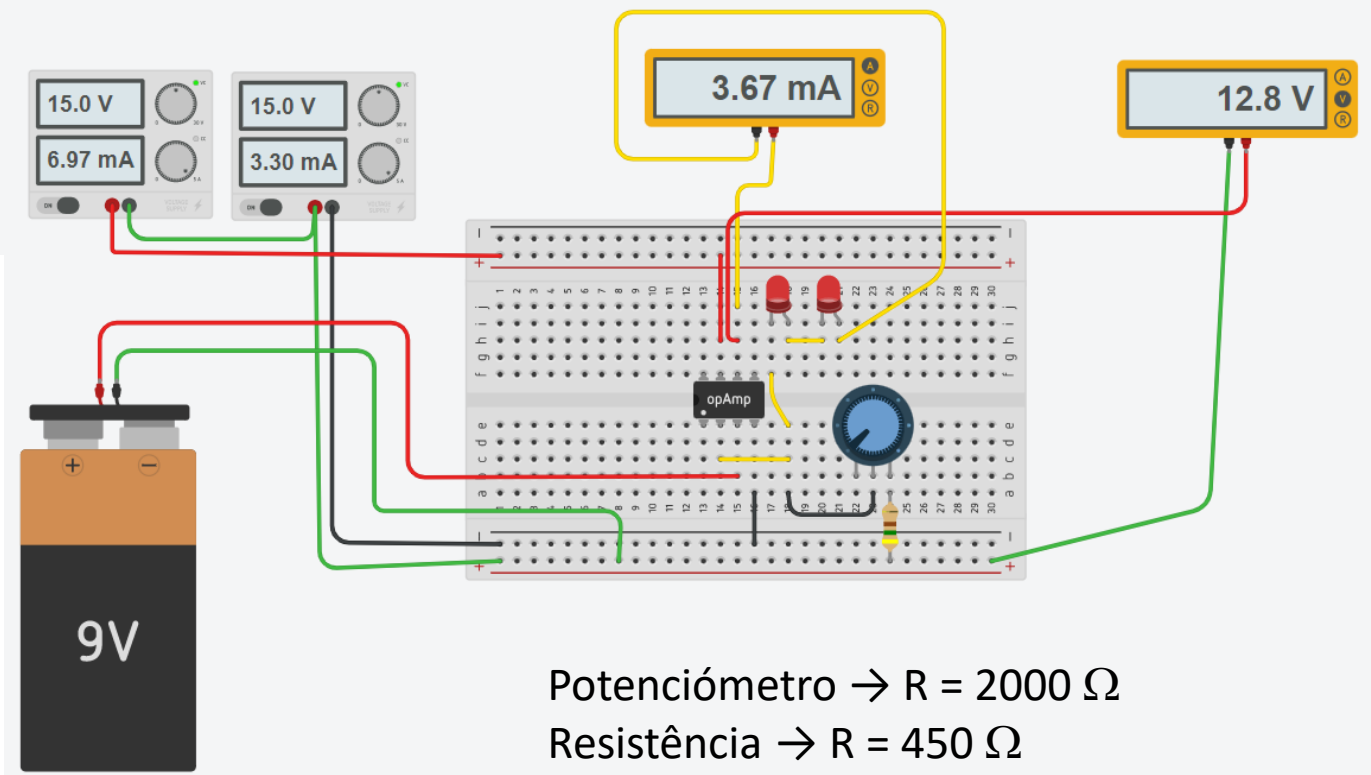
Conversor V→I (circuito iluminação LED)

https://www.tinkercad.com/things/a8hADgZhhhN?sharecode=GqLg8LT5Q0tusfbw2oU5cPzDK_P7DBEcxQJHjniB6xs



Potenciómetro → R = 0 Ω
Resistência → R = 450 Ω





Potenciómetro → R = 2000 Ω
Resistência → R = 450 Ω

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Conversor V→I versus Conversor I→V

$$R = \rho \times \frac{L}{S}$$



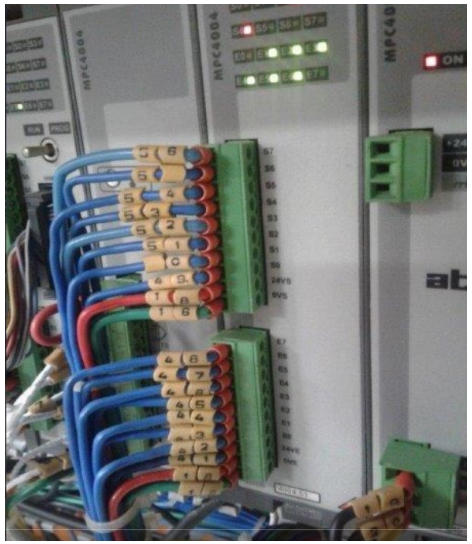
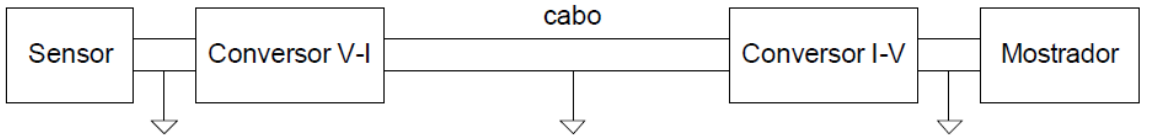
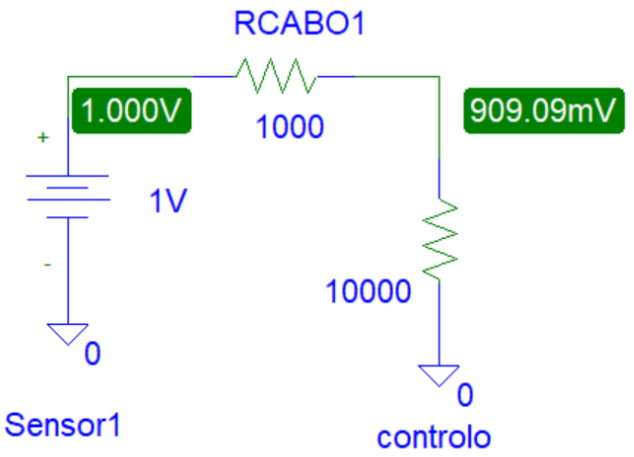


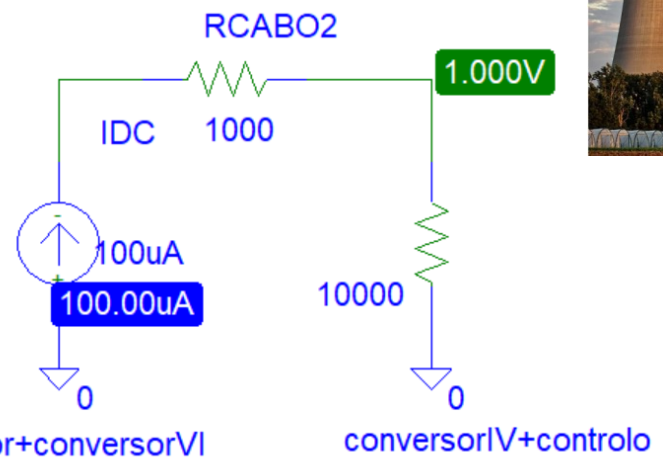


Figura 6.16. Aplicação industrial – circuitos conversores tensão-corrente e corrente-tensão



←[1]











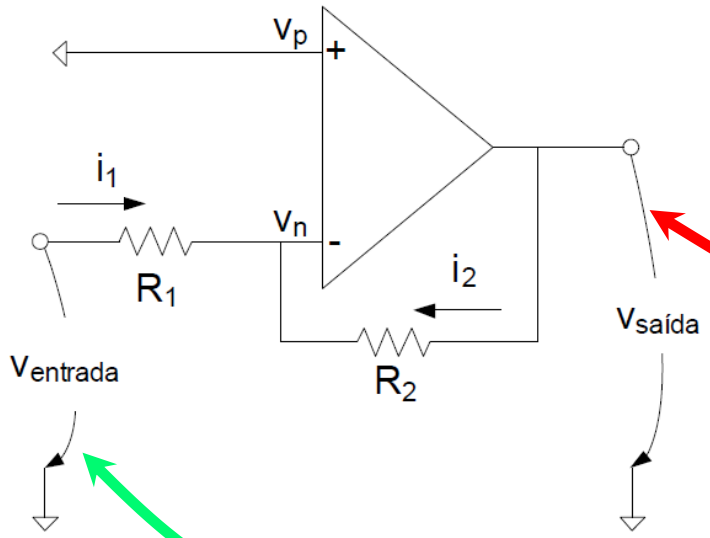


[1] Acácio Amaral, "Eletrónica Aplicada", Edições Sílabo, 2021.

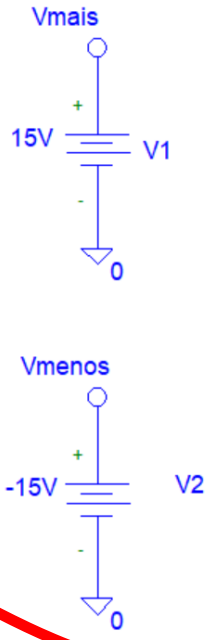
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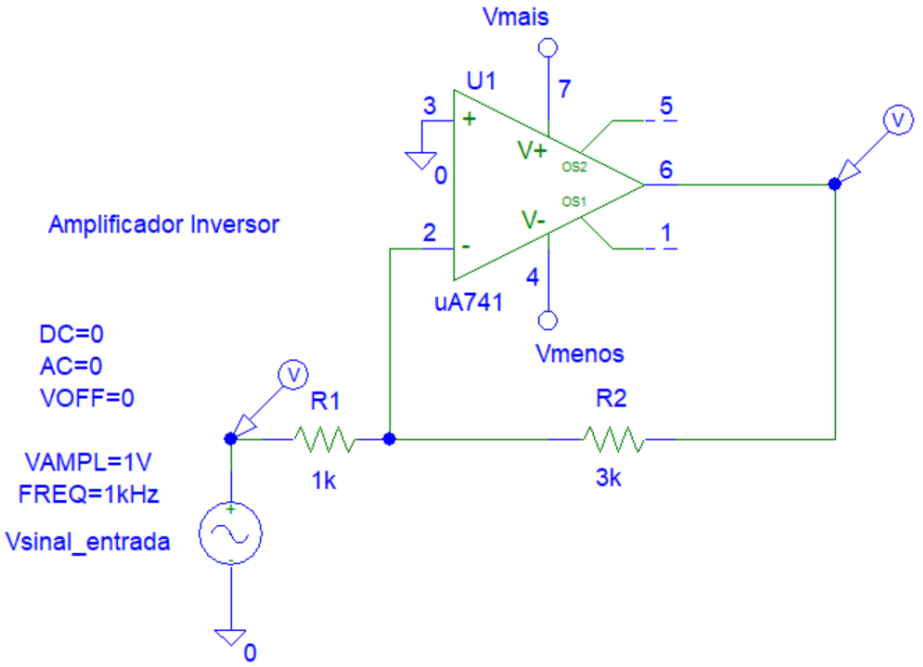
[1] Acácio Amaral, "Eletrónica Aplicada", Edições Sílabo, 2021.

Figura 6.20. Esquema elétrico do circuito amplificador inversor

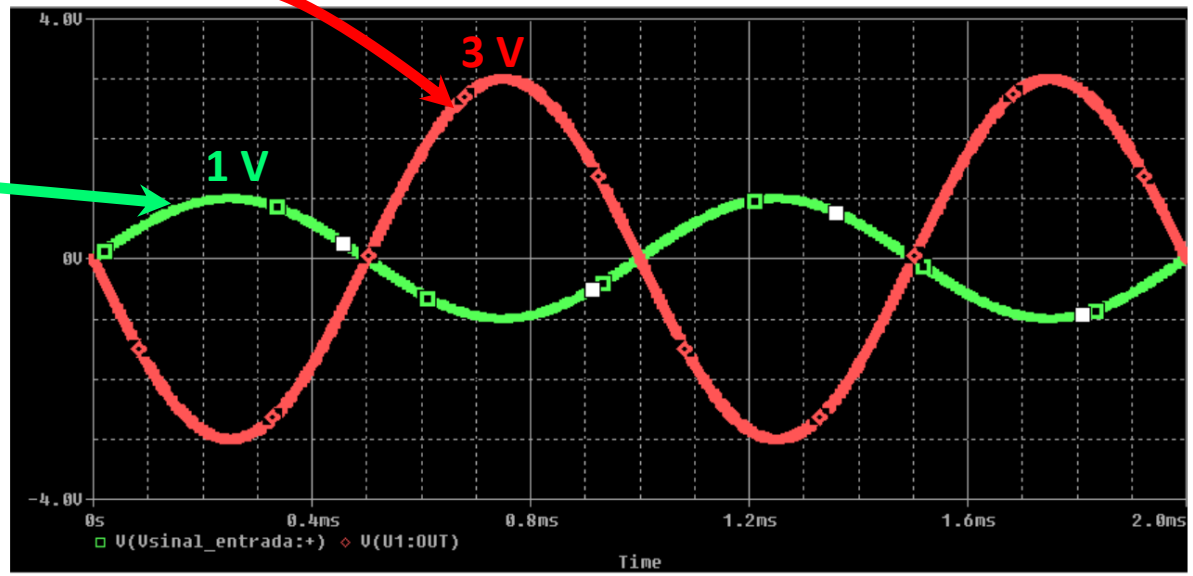


[1] →

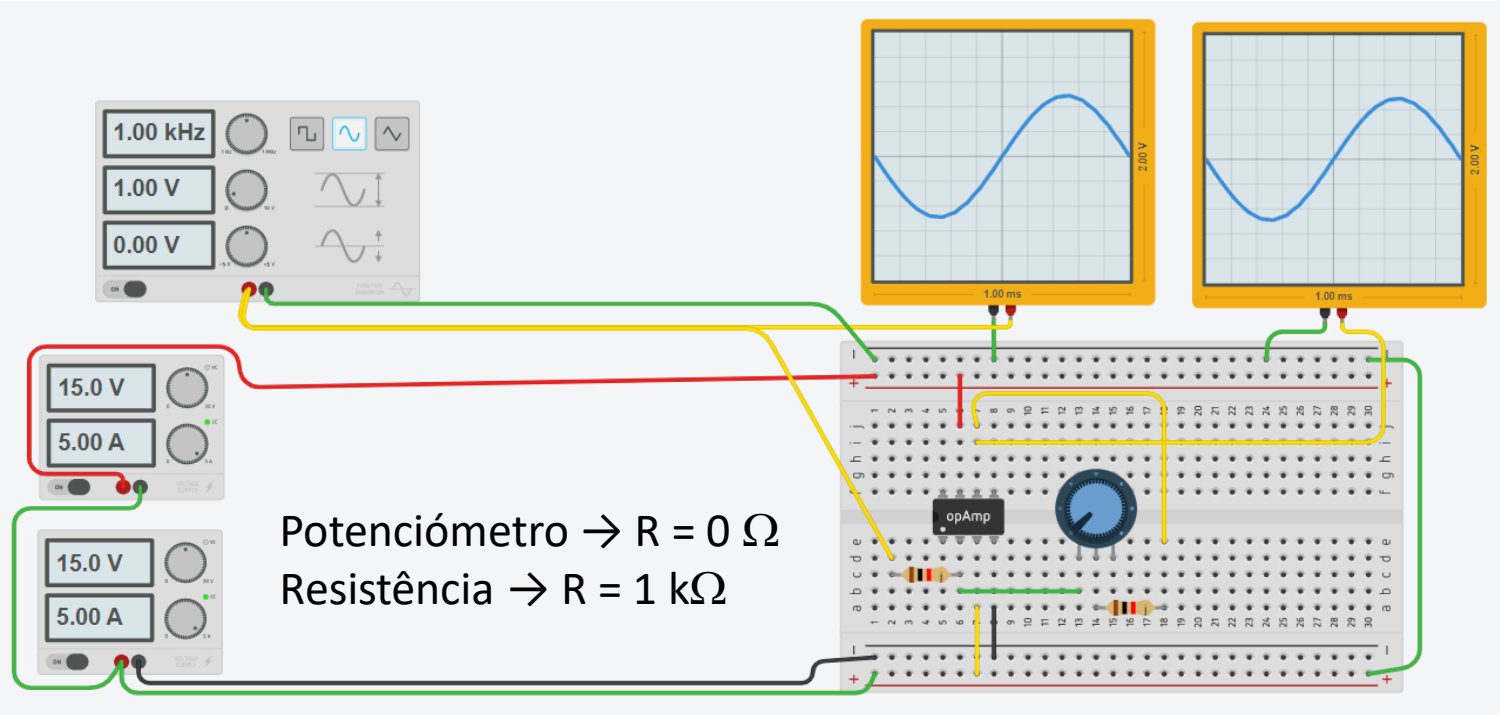




[1] → $$V_{saída} = -\frac{R_2}{R_1} \times V_{entrada} \qquad (6.34)$$



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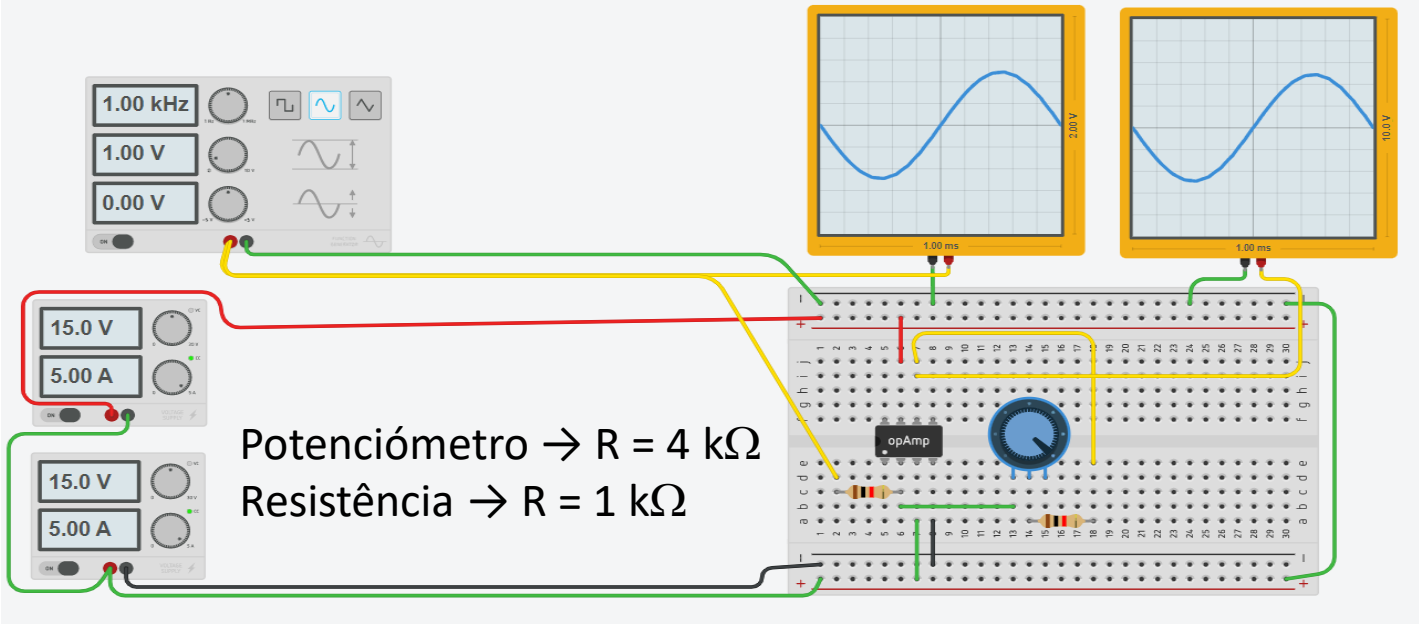


Amplificador inversor

Nota importante:

Saída e entrada desfasadas 180º

https://www.tinkercad.com/things/7VirrSG85l7?sharecode=Nqy0YQwkSZbLqxUFaanXkKkNQInqZq__ugmfEYW2XFs

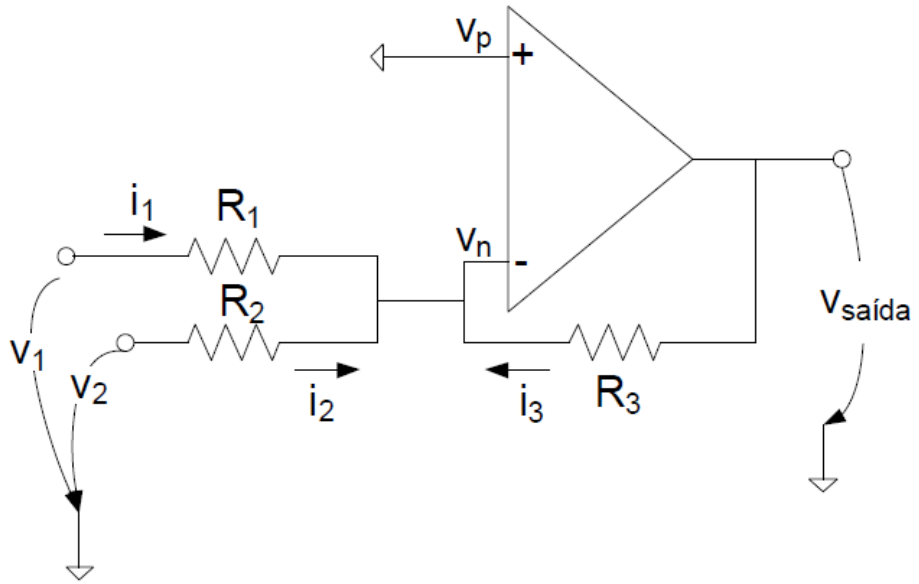


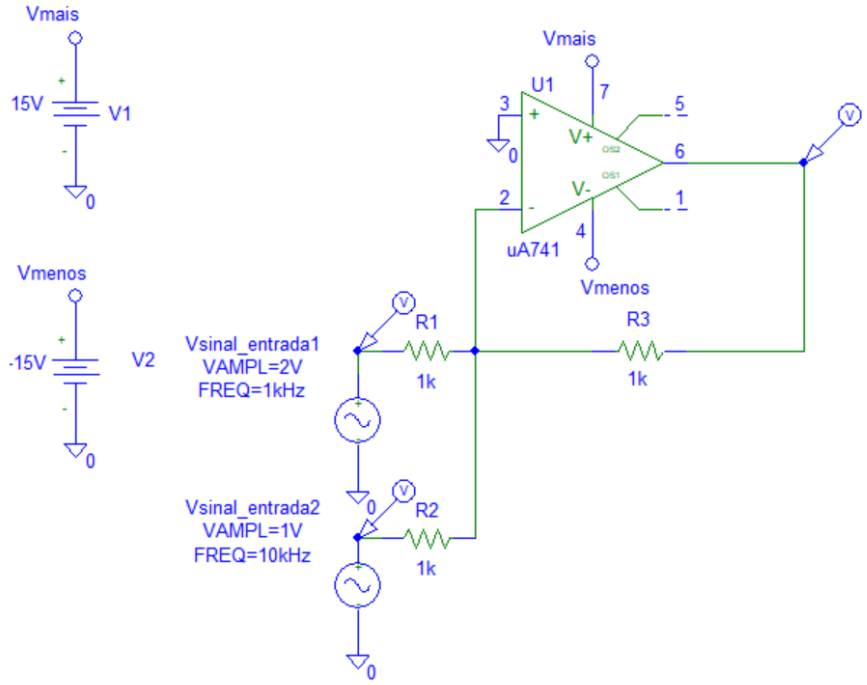
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[1] Acácio Amaral, “Eletrónica Aplicada”, Edições Sílabo, 2021.

Figura 6.24. Esquema elétrico de um somador de duas entradas

[1] →

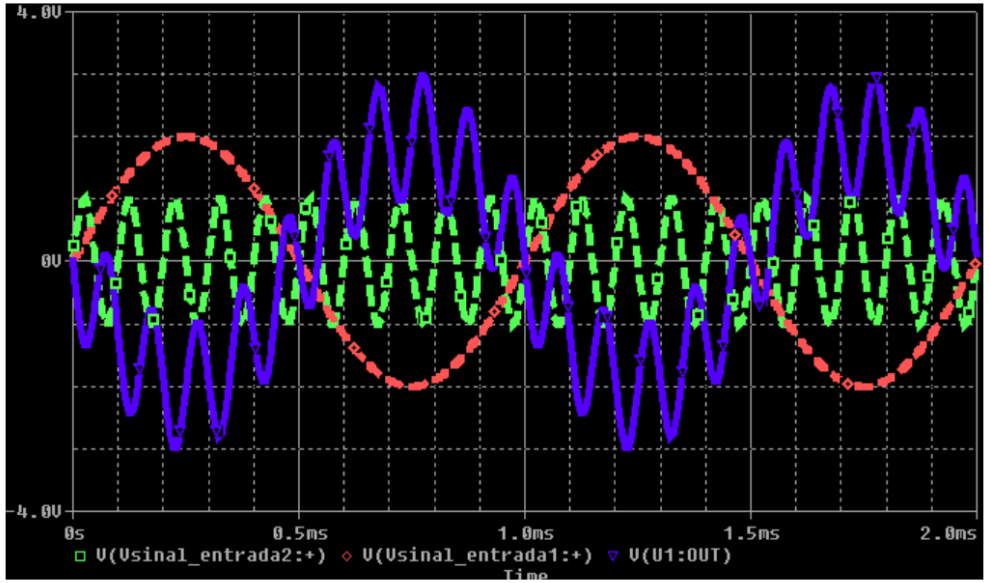




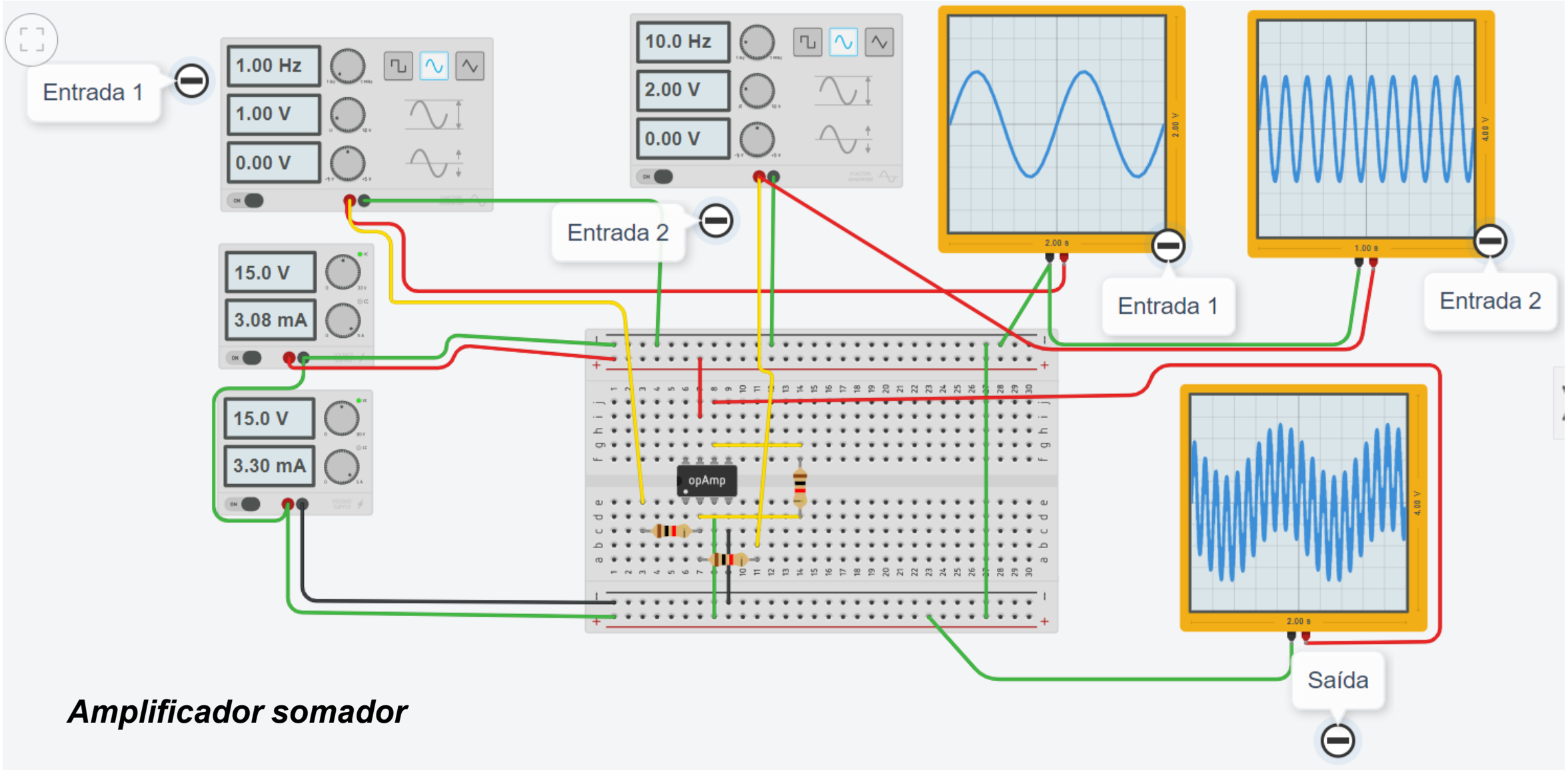
Amplificador somador

Se as resistências forem todas iguais ($R_1 = R_2 = R_3 = R$), então:

[1] → $$\frac{v_1}{R} + \frac{v_2}{R} + \frac{v_{saída}}{R} = 0 \quad \Rightarrow \quad v_{saída} = -(v_1 + v_2) \quad (6.41)$$



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Amplificador somador

https://www.tinkercad.com/things/6SHNvXTHuHk?sharecode=s3Z-UBz6nMZKUEBKjs31aYniA2ya3r8TP5B0pr8PiV0

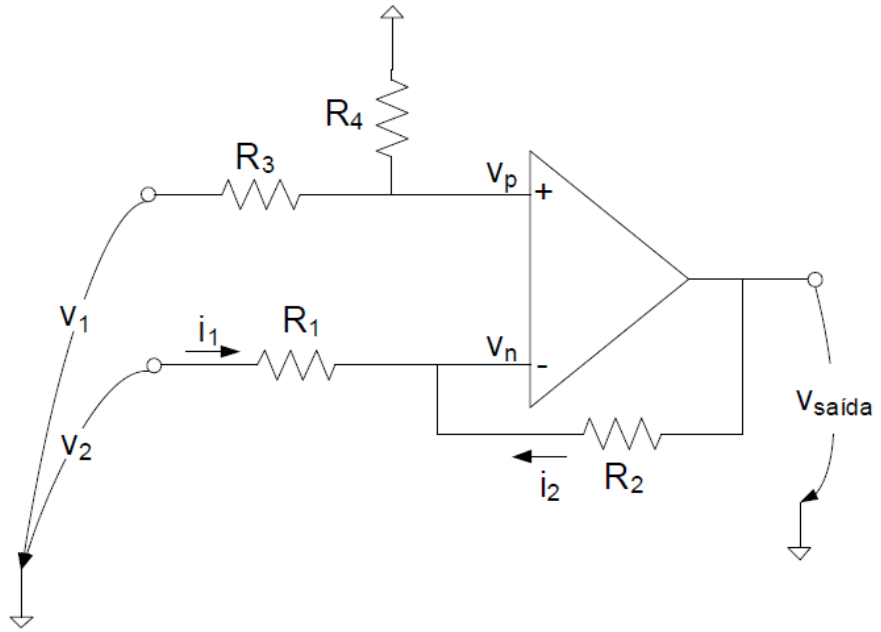
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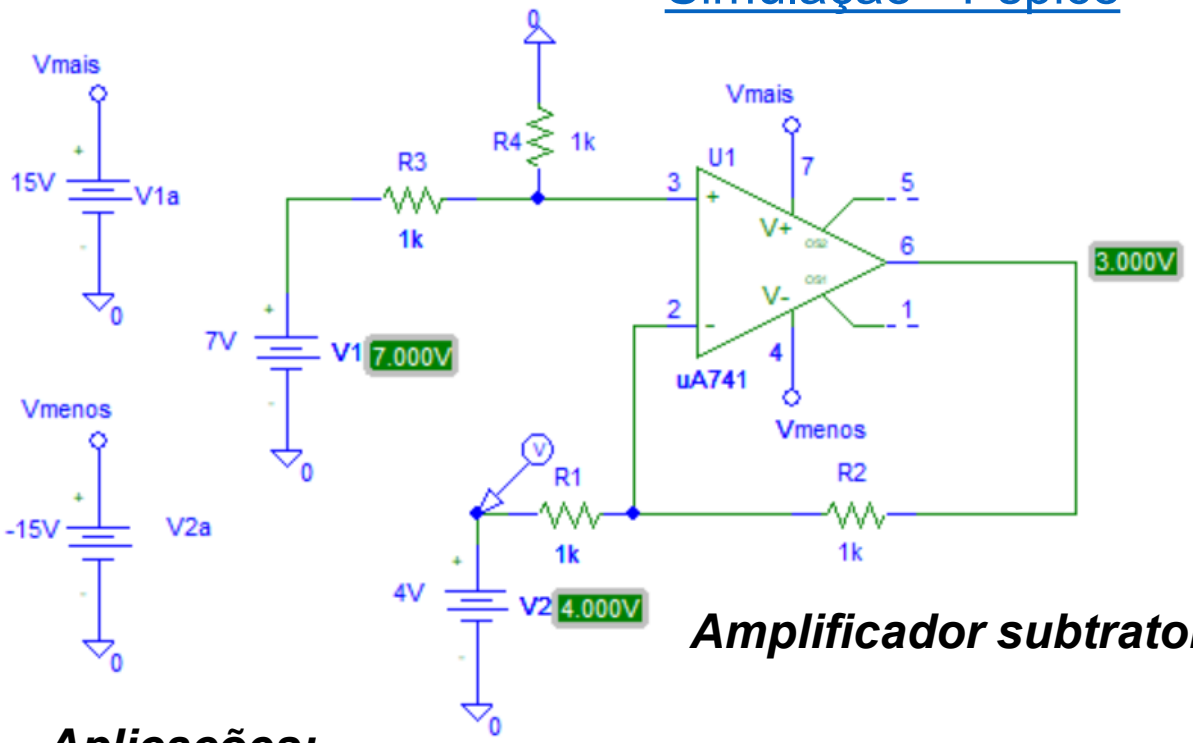
[1] Acácio Amaral, "Eletrónica Aplicada", Edições Sílabo, 2021.

Simulação - Pspice

Figura 6.29. Esquema elétrico de um subtrator

[1] →



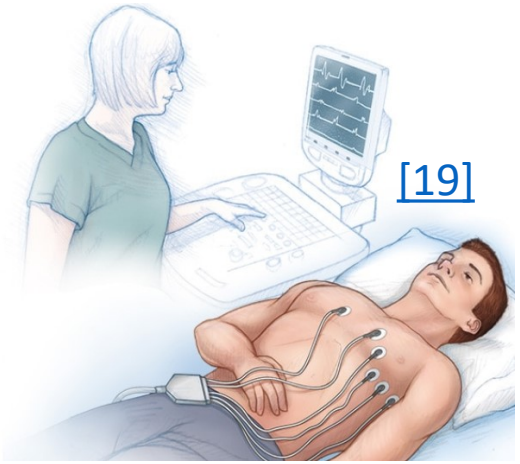


Amplificador subtrator

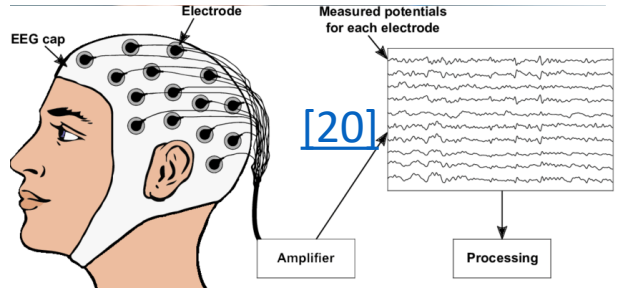
[1] →

$$\text{Se } R_1 = R_2 = R_3 = R_4 \qquad V_{saída} = V_1 - V_2 \qquad (6.51)$$

Aplicações:

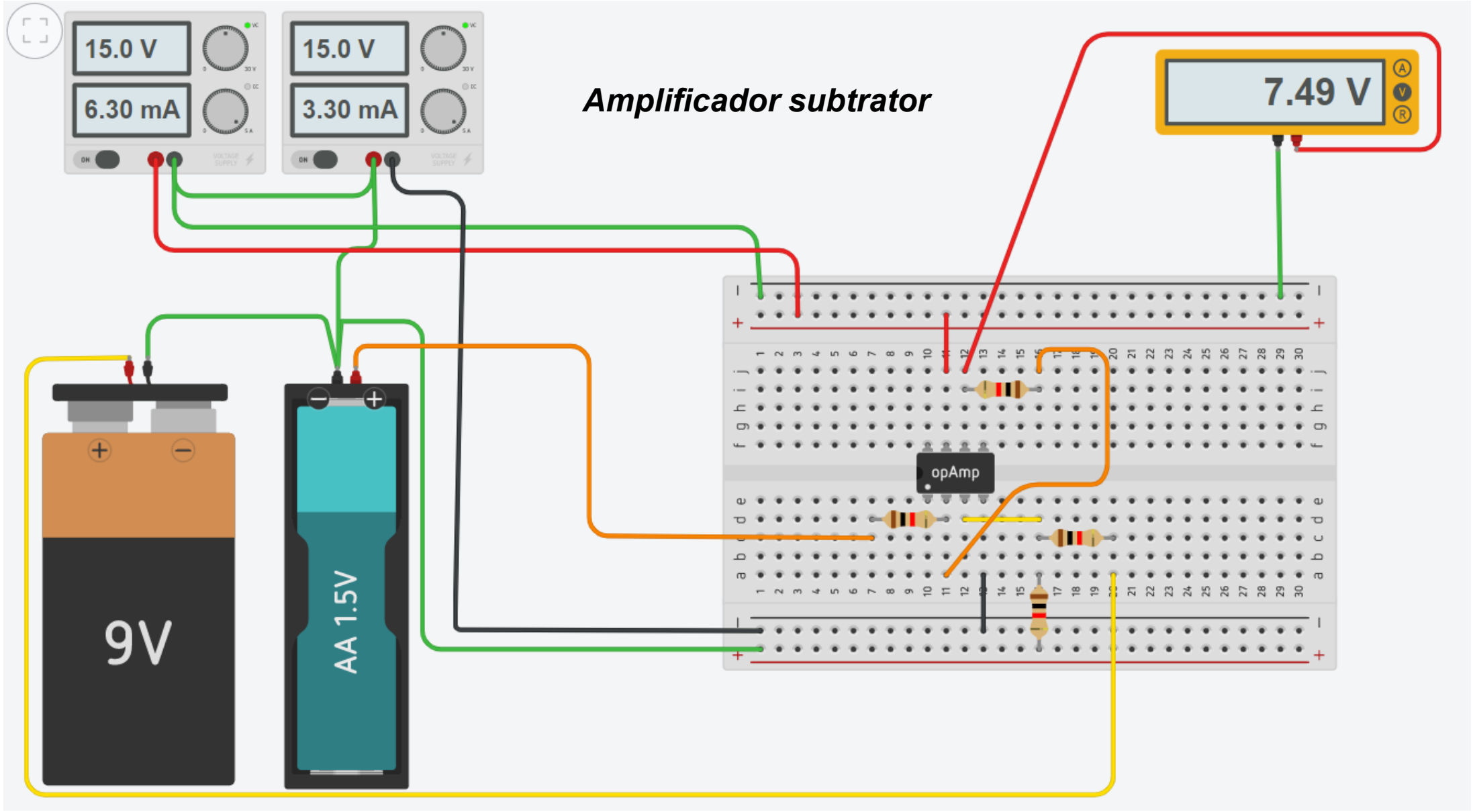


[19]



[20]

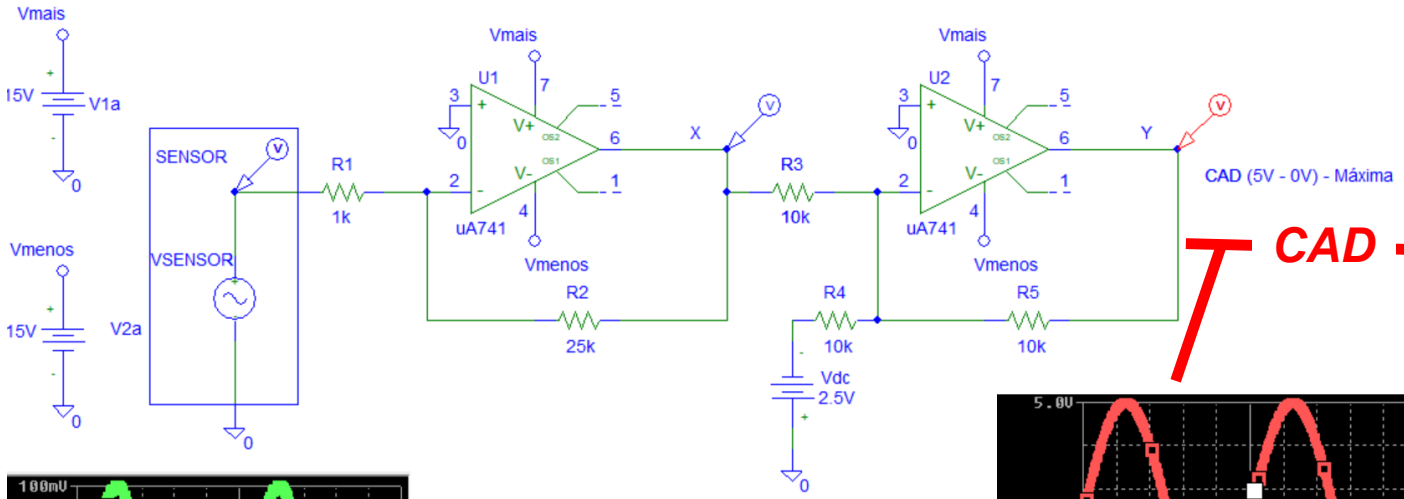
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https://www.tinkercad.com/things/9t4QFsXnCBG?sharecode=2_4Xc40sHZyXJUQrI69eySeWszQL-vPGt4RLVar7dPE

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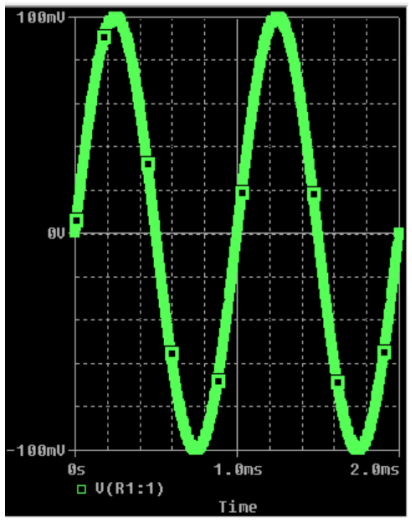
Adaptar níveis de tensão → Sensor (? V) → Circuito acondicionamento de sinal → CAD (5V)



CAD

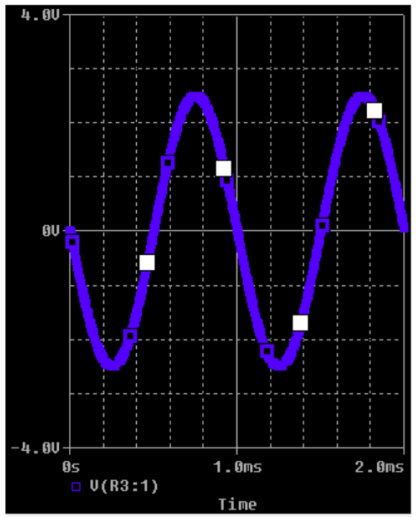


Computador analógico
Circuito de acondicionamento de sinal



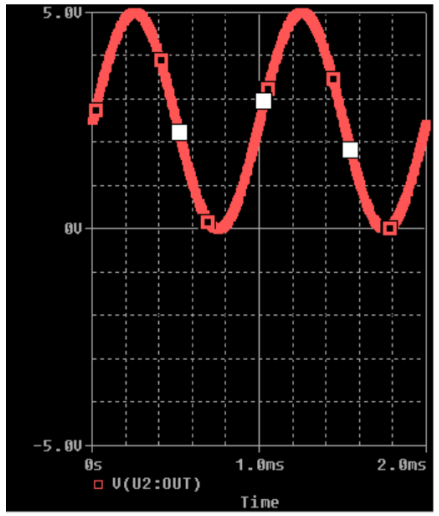
-0.1 V a 0.1 V

saída sensor



-2.5 V a 2.5 V

Amplificador inversor - 25 x



0 V a 5 V

somador inversor



[1] Acácio Amaral, "Eletrónica Aplicada", Edições Sílabo, 2021.

Exemplo: Quantização (3 bits) → 8 valores:
{0, 0.625, 1.25, 1.875, 2.5, 3.125, 3.75, 4.375}

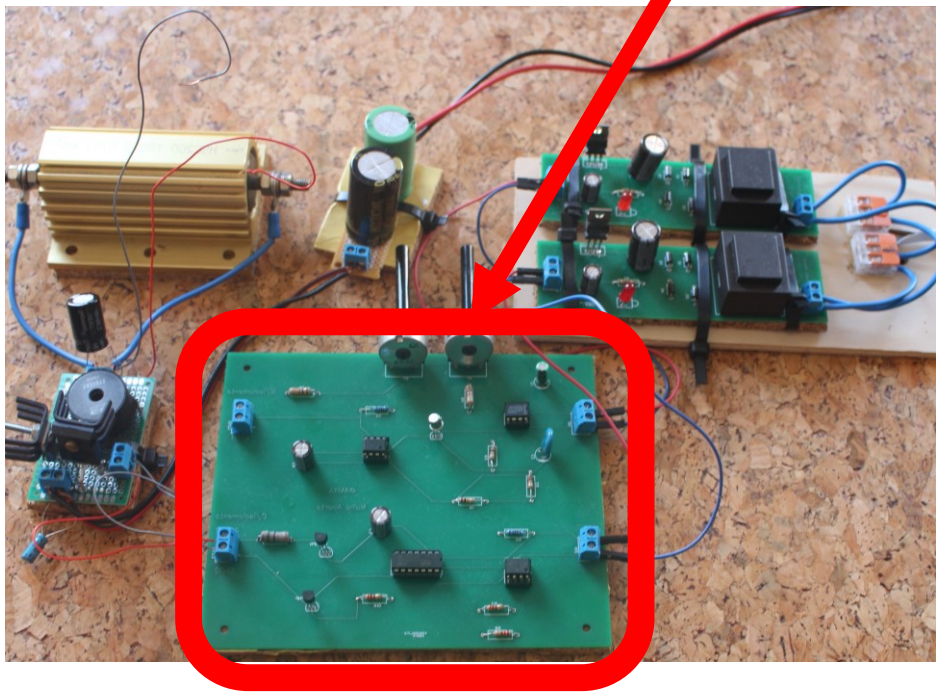
Processamento de Sinal Analógico
Deslocamento e Escalonamento de Sinais de Analógicos

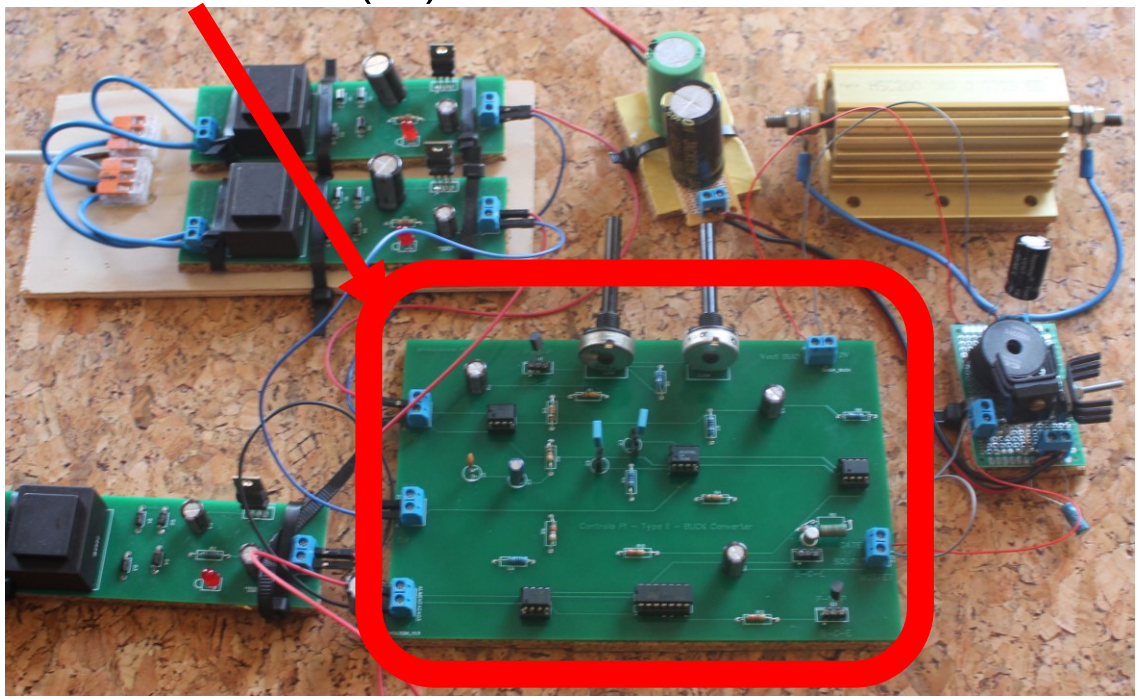
$$Sinal_{saida} = V_{SENSOR} \times 25 + V_{dc}$$

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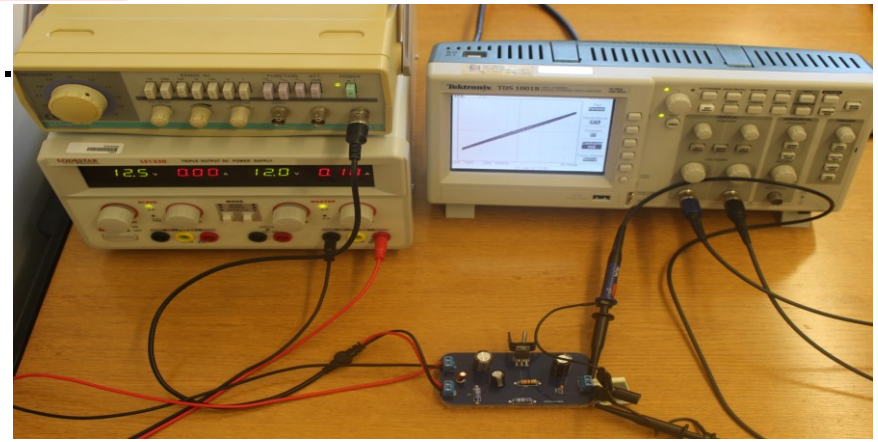
Aplicações [9, 20, 21]:

- Sistemas de controlo: malha aberta aberta[18] versus malha fechada (PI)[9]





- Sistemas de medida[19].



[9] Amaral, A. M. R., Cardoso, A. J. M. (Maio 2022). Using Python for the Simulation of a Closed Loop PI Controller for a Buck Converter. Signals, 3, pp. 313-325.

[20] Amaral, A. M. R. Cardoso, A. J. M. (May 2004) Use of ESR to predict failure of output filtering capacitors in boost converters. *IEEE International Symposium on Industrial Electronics*, pp. 1309-1314

[21] Amaral, A. M. R. , Cardoso, A. J. M. (Aug. 2009). A Simple Offline Technique for Evaluating the Condition of Aluminum–Electrolytic–Capacitors. *IEEE Transactions on Industrial Electronics*, 56(8), pp. 3230-3237.

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Aplicações [1]:

[1] Acácio Amaral, "Eletrónica Aplicada", Edições Sílabo, 2021.

[1]

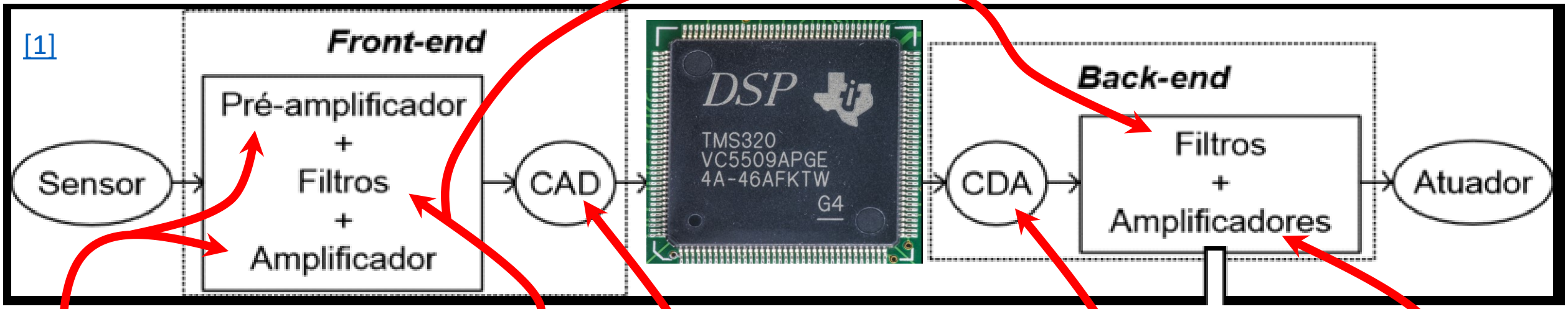


Figura 6.59. Filtro ativo do tipo passa-a-faixa [1]

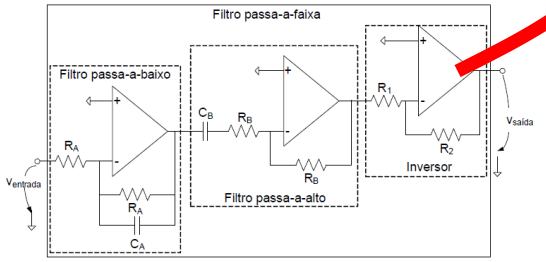


Figura 6.5. Circuito amplificador não inversor – cálculo da resistência de entrada em malha fechada [1]

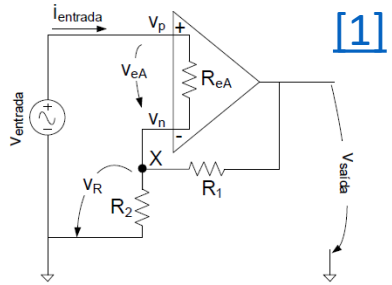


Figura 7.102. Composição de um CAD do tipo *flash* [1]

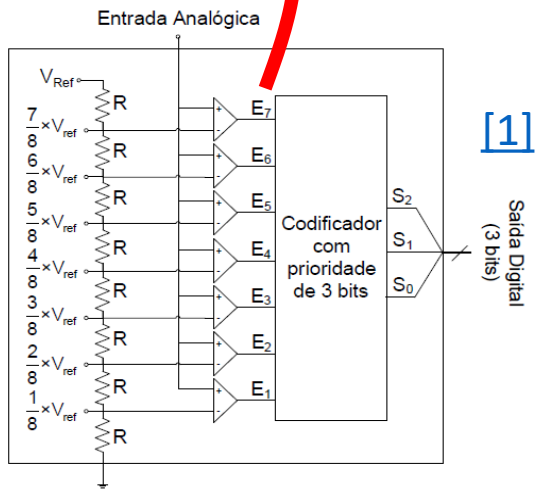


Figura 6.27. CDA de 3 *bits* e resolução de 1 Volt [1]

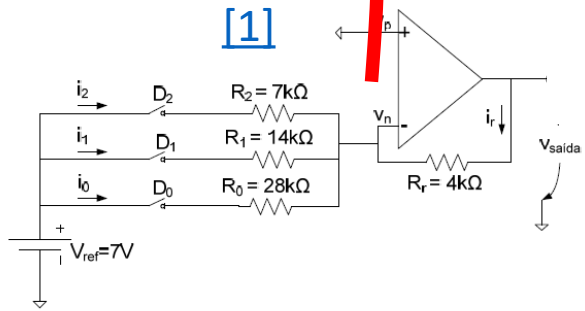
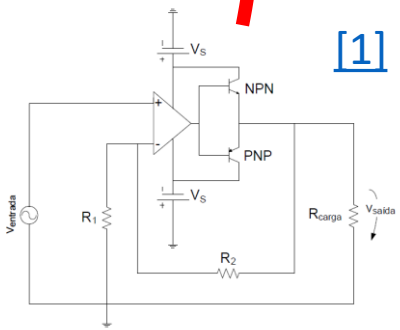


Figura 6.45. Amplificador classe *B* com um μA741 e dois transístores BJT interligados na configuração *push-pull* [1]



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